AI MEDICAL PRE-CONSULTATION SYSTEM

Qassim University
College of Computer | كلية الحاسب
جامعة القصيم

Faisal Abdullah Al-Nasyan — 431107944
Azzam Mohammed Al-Wahbi — 431107748
Mohammed Abdulrahman Al-Fawzan — 421109727
Saleh Zohir Al-Omair — 431109449
Mohammed Abdullah Al-Dkhily — 422113125

Supervisor: Dr. Fahad Ali Al-Oufi

INTRODUCTION

A bilingual web-based system that streamlines the pre-consultation process by collecting patient information and providing AI-powered preliminary diagnosis to physicians. The system enables patients to submit their medical data through traditional forms or voice interaction, while doctors receive AI-generated analysis before patient appointments, improving consultation quality and efficiency.

OBJECTIVE

To develop an intelligent pre-consultation system that automates patient data collection and provides preliminary medical analysis to physicians. The system aims to reduce consultation time, improve diagnostic accuracy, and enhance patient accessibility through voice-based interaction and bilingual support for both English and Arabic speakers.

METHODOLOGY

The system was developed using modern web technologies with a Next.js framework and PostgreSQL database. AI integration was implemented using OpenAI's language models for medical analysis and voice conversation capabilities. The architecture supports three user roles: patients who submit consultations, doctors who review reports, and administrators who manage the system.

IMPLEMENTATION TOOLS

DEVELOPMENT

NEXT.JS 15
REACT

DATABASE

NEON DB
DRIZZLE ORM

AI ANALYSIS

OpenAI
OPENAI GPT-5.5
OPENAI REALTIME

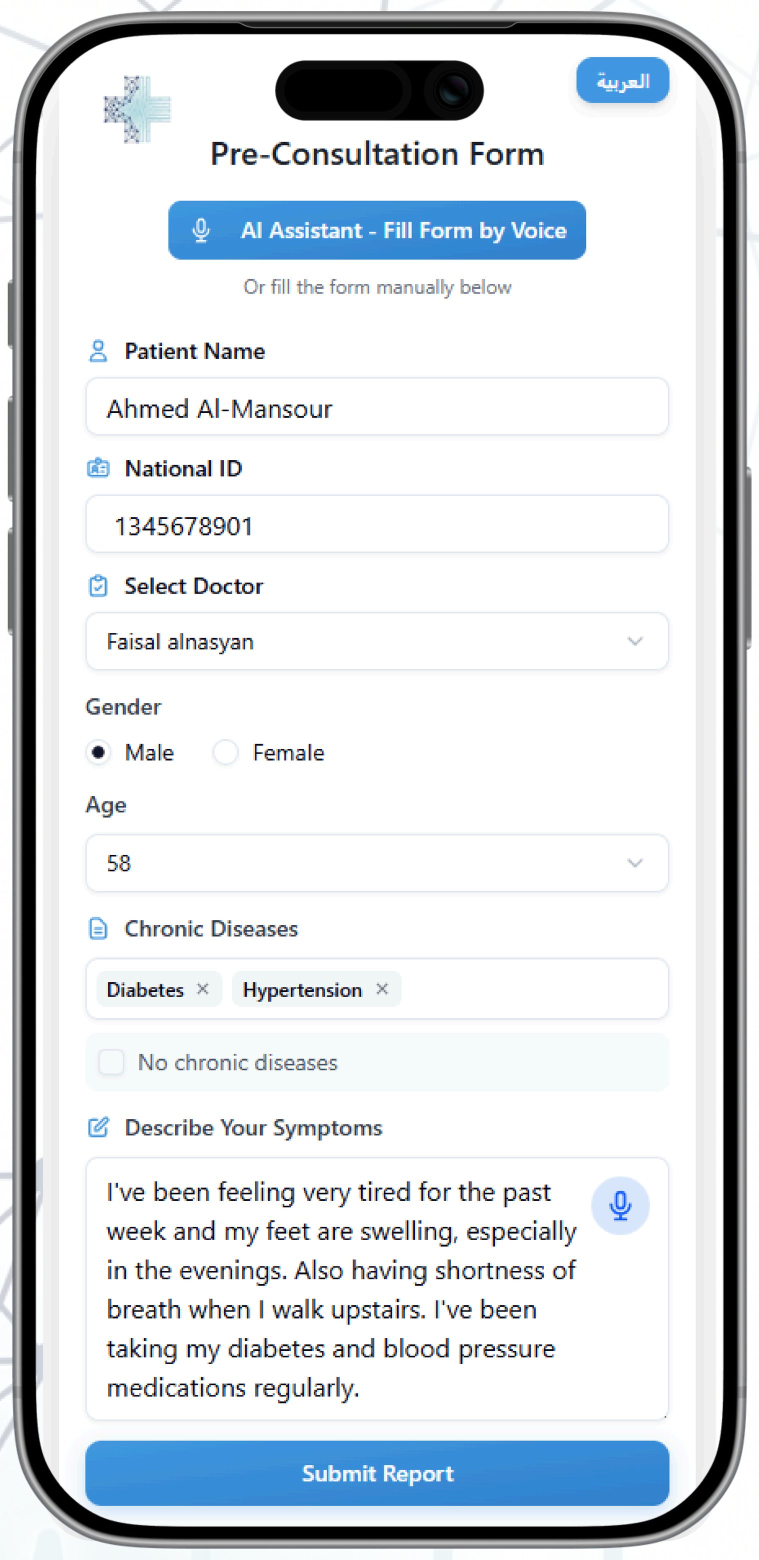
UI/UX

TAILWIND CSS
SHADCN/UI

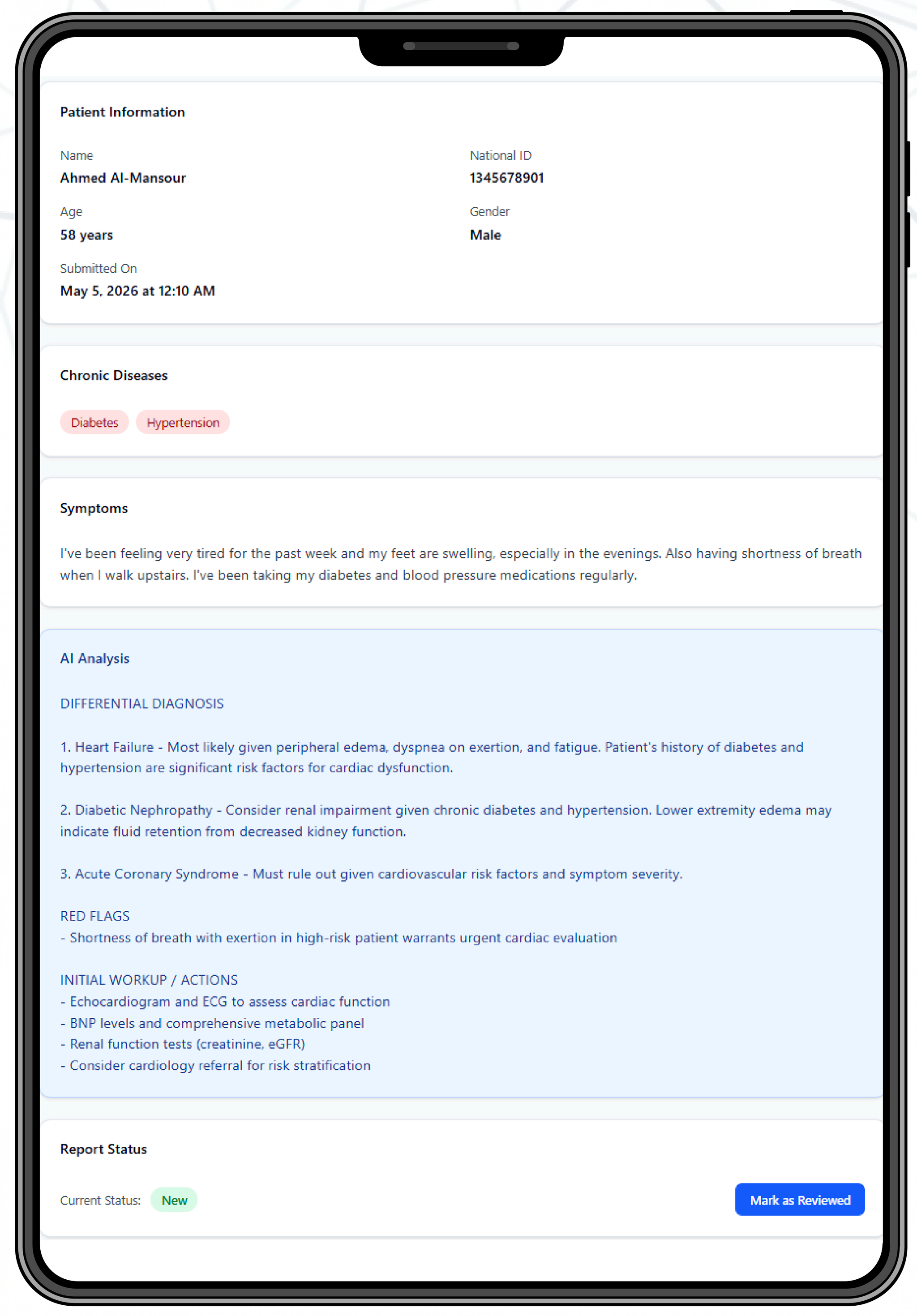
Tamheed

PROTOTYPE

PATIENT FORM



PATIENT REPORT



PROJECT IMPACT / KEY BENEFITS

FOR PATIENTS

Faster Access: Submit consultation requests anytime, anywhere without waiting in lines.

Voice Convenience: Fill forms by speaking naturally in English or Arabic.

Better Preparation: Patients can thoughtfully describe symptoms before seeing the doctor.

FOR DOCTOR

Time Efficiency: Review patient information and AI analysis before appointments.

Informed Decisions: Preliminary diagnosis suggestions help focus the consultation.

Reduced Workload: Automated data collection eliminates manual paperwork

CONCLUSION

The system successfully demonstrates how AI can enhance medical pre-consultation workflows. The integration of voice-based interaction significantly improved accessibility, while automated analysis provided valuable preliminary insights to physicians. The bilingual support ensures wider applicability in diverse healthcare settings.